Problem Statement:

Design a Yagi-Uda antenna with a gain of at least 10 dBi to receive television channel 43 (UHF) for a cable-ready television operating at 647 MHz.

Solutions:

Given Data:

Parameter	Value
Required Gain	≥ 10 dBi
Frequency	647 MHz
Input Impedance	75 Ω
Antenna Type	5-Element Yagi-Uda

Design Calculations:

1. Gain Selection

A 5-element Yagi-Uda antenna has a directivity of approximately 9.2 dBd.
Converting to dBi:
Gain = 9.2 + 2.15 = 11.35 dBi
Since 11.35 dBi > 10 dBi, the requirement is satisfied.

2. Mechanical Dimensions

Element Diameter = 1/4 inch = 0.635 cm
Boom Diameter = 5/8 inch = 1.587 cm

3. Wavelength Calculation

$$\lambda = \frac{C}{f} = \frac{3\times10^8}{647\times10^6}$$

$$\lambda = 0.4637 \text{ m} = 46.37 \text{ cm.}$$

Element Dimentions

Element	Formula	Length (cm)
Reflector	0.55 λ	25.50
Driven Element	0.50 λ	23.185
Director-1	0.45 λ	20.87
Director-2	0.40 λ	18.548
Director-3	0.35 λ	16.23

Element Spacing

Spacing	Formula	Distance (cm)
Reflector to Driven	0.35 λ	16.23
Driven to Director-1	0.125 λ	5.796
Director-1 to Director-2	0.20 λ	9.27
Director-2 to Director-3	0.20 λ	9.27

Design:

Conclusions

A 5-element Yagi-Uda antenna was successfully designed for 647 MHz operation. The calculated gain of 11.35 dBi exceeds the required 10 dBi, making the design suitable for receiving UHF television Channel 43.